

Supplementary Material: Structured Datasets, Methods, Evidence, and Protocols

Companion to: Person Re-Identification Datasets and Protocols: From Fixed-Source to Unified Multi-Source Retrieval

Companion repository:

<https://github.com/Trans1010/person-reid-datasets-methods-protocols>

This version accompanies the 88-reference manuscript and repository Release v1.2-paper-submission. References [1]–[88] follow that manuscript; [S1]–[S5] are supplementary-only sources. This supplementary material contains the detailed appendices referenced by the main manuscript. Numerical citations correspond to the main manuscript reference list; S-prefixed citations identify supplementary-only sources listed at the end of this document. The companion repository provides machine-readable registry files, evidence links, and protocol descriptions associated with this review. Restricted dataset files are not redistributed. Registry entries were last verified on 4 September 2026. The primary database search in Table S1 was completed on 4 August 2026, followed by an update search completed on 17 September 2026.

Table S1. Representative search strategies.

Database	Representative search strategy	Time range	Last search date
Web of Science Core Collection: core discovery	TS=("person re-identification" OR "person re-ID") AND TS=(dataset* OR benchmark* OR "evaluation protocol")	1 January 2007–4 August 2026	4 August 2026
IEEE Xplore: core discovery	("person re-identification" OR "person re-ID") AND (dataset OR benchmark OR "evaluation protocol")	1 January 2007–4 August 2026	4 August 2026

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Table S1 (continued)

Database	Representative search strategy	Time range	Last search date
Google Scholar: core discovery	("person re-identification" OR "person re-ID") combined with (dataset OR benchmark OR "evaluation protocol")	1 January 2007–4 August 2026	4 August 2026
Branch coverage and update checks: primary databases, Crossref, OpenAlex, arXiv, CVF Open Access and publisher pages	Core discovery query with each branch term added separately: infrared, multispectral, multimodal, cross-modal, UAV, drone, aerial, aerial-ground, mixed gallery, mixed-source, unified retrieval, any-scenario, video, text query, clothes-changing, lifelong and open-set	5 August 2026–17 September 2026	17 September 2026

Searches were conducted in two stages. Core discovery queries identified candidate person ReID dataset, benchmark and protocol papers without requiring a branch term. Branch-coverage checks then added each branch term separately. Early ground RGB benchmarks were also confirmed through citation tracking, recent reviews and method papers. These procedures supported purposive selection and were not used to claim exhaustive coverage.

Table S2. Comparability rules for ReID evaluation results.

Comparison type	Direct Rank-1/mAP comparison?	Required controls	Valid conclusion	Unsupported claim
Same dataset, same split, same direction	Yes	Identical splits, pedestrian bounding boxes, gallery filtering, training data and number of models	Assess the relative gains from the added mechanism in this setting	Cross-direction or cross-protocol capability
Same dataset, different protocols	Conditional	Specify the query direction, gallery size, source composition and filtering rules	Compare the relative performance drop caused by changes in settings; direct ranking is not possible	Absolute performance ranking across protocols

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Table S2 (continued)

Comparison type	Direct Rank-1/mAP comparison?	Required controls	Valid conclusion	Unsupported claim
Opposite query directions	No	Report A→B and B→A separately, and control query/gallery sizes	Analyse directional asymmetry	Automatic equivalence of the reverse task
Separately trained vs unified model	No	Indicate parameter sharing, task heads, routing, scene labels and the number of training runs	Assess the performance and resource costs of unified training	Fixed source pairs as a substitute for a mixed gallery
Cross-dataset comparison	No	Control training sources, pretraining and target-data use only within explicitly defined transfer settings	Compare task coverage or cross-domain generalization, without ranking absolute performance	Absolute performance ranking across datasets

This table provides the detailed controls behind the comparability principle stated in Section 6.1 of the main text.

Appendix A Structured master table of representative datasets

The table below ensures that data fields outside the main narrative remain traceable. Dataset sizes and acquisition conditions are verified preferentially against the original dataset papers, official project pages or formal releases. Information not explicitly reported in the listed evidence sources uses the label ‘Not explicitly reported in the listed evidence sources (checked 2026-09-04).’—No value is inferred.

Table A1. Structured master table of representative datasets. Panel A of 2; the identifier column is repeated.

Dataset	Role	IDs / Images	Modality	Platform
VIPeR [15]	Ground RGB core	632 / 1,264	RGB	Ground
CUHK03 [16, 17]	Detection-error bridge	1,360 / 13,164 Original; 1,467 / 14,096 NP	RGB	Ground
Market-1501 [18]	Large-gallery core	1,501 / 32,668	RGB	Ground

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Table A1 (continued), panel A

Dataset	Role	IDs / Images	Modality	Platform
MSMT17 [19]	Multi-environment RGB core	4,101 / 126,441	RGB	Ground
RegDB [22]	Cross-spectrum branch	412 / 8,240	RGB, TIR	Ground
SYSU-MM01 [23]	Multi-camera cross-spectrum	491 / raw: 287,628 RGB + 15,792 IR; commonly used filtered evaluation count: 30,071 RGB + 15,792 IR = 45,863	RGB, IR	Ground (4 RGB + 2 IR)
RGBNT201 [24]	Aligned tri-modal bridge	201 / 14,361; 4,787 triplets	RGB, NIR, TIR	Ground
PRAI-1581 [25]	UAV branch	1,581 / 39,461	RGB	UAV (2 DJI; 20–30 m)
AG-ReID.v2 [27]	Aerial-ground cross-platform	1,615 / 100,502	RGB	UAV, CCTV, wearable
CARGO [14]	Synthetic aerial-ground bridge	5,000 / 108,563	RGB	5 aerial + 8 ground
CP2108 [4]	Time and environment combination	2,108 / 142,817	RGB	17 UAV + 22 ground
AT-USTC [28]	Spectrum and long-term clothing	270 / 135,465	RGB, IR	8 RGB + 8 IR cameras
MP-ReID [3]	Multimodal, multiplatform fixed settings	1,930 / 136,156	RGB, IR, TIR	Ground + UAV
WHU-MARS [5]	Heterogeneous mixed-gallery	2,337 / 434,620	RGB, NIR, TIR	UAV, Ground
CUHK01 [S1]	Early ground RGB	971 / 3,884	RGB	Ground
CUHK02 [S2]	Early ground RGB	1,816 / 7,264	RGB	Ground
GRID [42]	Gallery-distractor milestone	250 matched IDs + distractors	RGB	Ground
DukeMTMC-reID [S3, S4]	ReID derivative of DukeMTMC; large ground-RGB benchmark	1,404 / 36,411	RGB	Ground
MARS [29]	Video ReID milestone	1,261 / 20,478 tracklets	RGB	Ground
iLIDS-VID [30]	Early video ReID	300 / 600 tracklets	RGB	Ground
LLCM [53]	Low-light VI branch	1,064 / 46,767 boxes	RGB, IR	Ground

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Table A1 (continued), panel A

Dataset	Role	IDs / Images	Modality	Platform
HITSZ-VC [31]	VI $\times$ video combination	927 / 21,863 tracklets	RGB, IR	Ground
PRCC [84]	Clothes-changing bridge	221 / 33,698	RGB	Ground
LTCC [82]	Long-term clothing change	152 / 17,138	RGB	Ground
VC-Clothes [S5]	Synthetic clothing change	512 / 19,060	RGB	Ground
DeepChange [83]	Long-term clothing benchmark	1,121 / 178,407 boxes	RGB	Ground
AG-ReID.v1 [61]	Early aerial-ground version	388 / 21,983	RGB	UAV, Ground
AG-VPreID [32]	Aerial-ground video combination	6,632 / 32,321 tracklets / 9.6M+ frames	RGB	Drone, CCTV, wearable
CUHK-PEDES [34]	Text-query branch	13,003 / 40,206 images / 80,412 texts	RGB + text	Ground
RSTPreID [35]	Text-query bridge	4,101 / 20,505 images	RGB + text	Ground

Table A1. Structured master table of representative datasets. Panel B of 2; the identifier column is repeated.

Dataset	Time / Clothing	Unit	Main Query-Gallery
VIPeR [15]	Short-term	Image	Two-camera cross-view; 2008 representative paper
CUHK03 [16, 17]	Short-term	Image	Manual/detected boxes; Original and NP protocols
Market-1501 [18]	Short-term	Image	Detected boxes; distractor gallery
MSMT17 [19]	Multiple time periods	Image	Multi-camera RGB retrieval
RegDB [22]	Short-term	Image	Visible $\leftrightarrow$ Thermal; two directions
SYSU-MM01 [23]	Indoor/outdoor; single/multi-shot	Image	All-search / Indoor-search; filtered subset separated from raw collection
RGBNT201 [24]	Synchronized capture	Image	Aligned RGB-NIR-TIR matching
PRAI-1581 [25]	Short-term; $\sim$ 60 videos	Image	UAV intra-platform retrieval; formal 2021 volume; Early Access 2020
AG-ReID.v2 [27]	Short-term; 15 soft attributes	Image	A $\rightarrow$ C, A $\rightarrow$ W, C $\rightarrow$ A, W $\rightarrow$ A
CARGO [14]	Synthetic; controllable	Image	ALL; G $\leftrightarrow$ G; A $\leftrightarrow$ A; A $\leftrightarrow$ G

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Table A1 (continued), panel B

Dataset	Time / Clothing	Unit	Main Query–Gallery
CP2108 [4]	Seasons, day/night, partial clothing change	Image	ALL mixed retrieval; A→G and G→A separately
AT-USTC [28]	One year; 710 clothing sets	Image	Six Anytime-ReID scenarios
MP-ReID [3]	Short-term	Image	Six pair experiments, twelve directions; pair-specific splits
WHU-MARS [5]	Seasons, day/night	Image	AS-ReID: known-source heterogeneous mixed gallery
CUHK01 [S1]	Short-term	Image	Cross-camera matching
CUHK02 [S2]	Short-term	Image	Cross-camera matching
GRID [42]	Short-term	Image	Matched pairs with distractor gallery
DukeMTMC-reID [S3, S4]	Short-term	Image	702 train / 702 test IDs; 2,228 query; 17,661 gallery; detected-box multi-camera retrieval
MARS [29]	Short-term	Tracklet	Video query–gallery retrieval
iLIDS-VID [30]	Short-term	Tracklet	Cross-camera video retrieval
LLCM [53]	Low-light	Image	Visible↔Infrared low-light retrieval
HITSZ-VCM [31]	Video; low-light	Tracklet	Visible–infrared video retrieval
PRCC [84]	Clothing change	Image	Same-clothes and clothes-changing protocols
LTCC [82]	Two months; clothing change	Image	Long-term clothes-changing retrieval
VC-Clothes [S5]	Synthetic clothing change	Image	Synthetic cross-clothes retrieval
DeepChange [83]	12 months; clothing change	Image	Long-term cross-clothes retrieval
AG-ReID.v1 [61]	Short-term	Image	Aerial↔Ground
AG-VPreID [32]	Video; multi-platform	Tracklet	Aerial–ground video retrieval
CUHK-PEDES [34]	Short-term	Image + text	Text→image person retrieval
RSTPreid [35]	Short-term	Image + text	Text→image person retrieval

For datasets such as CUHK03 and SYSU-MM01, the original statistics differ from those of commonly used evaluation versions. Table A1 retains version notes at the data-content level. Specific training–test splits, query–gallery construction, retrieval directions, gallery mixing status and filtering rules are recorded consistently in Appendix E. The same principle distinguishes data content from evaluation protocols for

multi-source datasets such as MP-ReID and WHU-MARS.

Appendix B Representative Method–Evidence Registry

Table B1 records representative methods’ original design objectives, original benchmarks, subsequent re-evaluation settings, parameter-sharing strategies, use of scene labels and direct evidence boundaries. Baseline results in subsequent benchmarks do not change the tasks for which the methods were originally designed.

Table B1. Representative Method–Evidence Registry. Panel A of 2; the identifier column is repeated.

Method	Original design target	Original benchmarks	Later re-evaluated on	Unified/shared model?
XQDA [41]	Ground RGB	VIPeR, Market-1501	—	No
PCB [47]	Ground RGB	Market-1501	—	No
BoT [45]	Ground RGB	Market-1501, MSMT17	WHU-MARS AS-ReID	Later baseline only
AGW [1]	Ground RGB	Market-1501, MSMT17	WHU-MARS AS-ReID	Later baseline only
TransReID [49]	Ground RGB	Market-1501, MSMT17	MP-ReID, WHU-MARS	Shared backbone; later tests
CAJ [51]	VI-ReID	SYSU-MM01, RegDB	MP-ReID	Pair-specific
DEEN [53]	VI-ReID	SYSU-MM01, LLCM	MP-ReID	Pair-specific
SAAI [54]	VI-ReID	SYSU-MM01	MP-ReID	Pair-specific
CSDN [56]	VI-ReID	SYSU-MM01	MP-ReID	Pair-specific
RotTrans [60]	UAV RGB	PRAI-1581	—	No
VDT [14]	Aerial–ground	CARGO, AG-ReID	CP2108	Pair-specific
VIF-AGReID [62]	Aerial–ground	AG-ReID, CARGO	—	Pair-specific
ViSA [63]	Aerial–ground	AG-ReID, CARGO, LAGPeR	—	Pair-specific
SVPR-ReID [4]	Aerial–ground	CP2108	—	Pair-specific
Uni-Prompt [3]	Multimodal–multiplatform fixed settings	MP-ReID	—	One model across fixed pairs
UAD [5]	Heterogeneous mixed gallery	WHU-MARS	—	Single model

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Table B1 (continued), panel A

Method	Original design target	Original benchmarks	Later re-evaluated on	Unified/shared model?
CLIP-ReID [65]	Ground RGB / vision-language	Market-1501	MP-ReID, WHU-MARS	Later baseline only

Table B1. Representative Method–Evidence Registry. Panel B of 2; the identifier column is repeated.

Method	Train scenario labels	Test scenario labels	Direct evidence	Evidence boundary
XQDA [41]	No	No	Ground RGB metric learning	No cross-spectrum/platform evidence
PCB [47]	No	No	Ground RGB local parts	No mixed-gallery evidence
BoT [45]	No	No	Direct on original and later benchmark	Later evaluation does not change original target
AGW [1]	No	No	Ground RGB and later mixed-gallery baseline	No original mixed-source design
TransReID [49]	Protocol-dependent	Protocol-dependent	Direct on listed protocols	Original design remains Ground RGB
CAJ [51]	Yes	Yes	Visible–infrared protocols	No direct mixed-gallery evidence
DEEN [53]	Yes	Yes	VI and low-light protocols	No direct platform-combination evidence
SAAI [54]	Yes	Yes	Identity-level cross-spectrum alignment	No mixed-gallery evidence
CSDN [56]	Yes	Yes	CLIP-assisted VI retrieval	No direct open-identity evidence
RotTrans [60]	No	No	UAV intra-platform	No aerial–ground evidence
VDT [14]	Yes	No	Original aerial–ground protocols; later CP2108 baseline	Later CP2108/ALL evidence does not change original design target
VIF-AGReID [62]	Yes	No	Aerial–ground cross-platform	No cross-spectrum mixed-gallery evidence
ViSA [63]	Yes	Yes	Viewpoint-aware aerial–ground	No proof of universal mixed-gallery ability
SVPR-ReID [4]	Yes	No	Semantic aerial–ground protocols	No open-identity evidence
Uni-Prompt [3]	Yes	Yes	Six pair experiments	Fixed directions are not mixed gallery
UAD [5]	Yes	Yes	AS-ReID unified ranking	Known-source closed-world scope
CLIP-ReID [65]	No	No	Original and later baseline results	Text pretraining is not text-query evidence

Appendix C Data availability and ethics fields

Table C1 records availability, dataset licences or agreements, ethics/informed consent, anonymisation, usage restrictions, maintenance status and availability-check dates for each dataset. It also adds three columns: Evidence source, Official URL or archived URL, and Source version/date. The Evidence source column lists original papers, official project pages, official repositories, release/usage agreements or archived sources that support the field assessments. The URL column records the corresponding access points. Information not explicitly reported in these sources is uniformly marked as ‘Not explicitly reported in the listed evidence sources (checked 2026-09-04).’—This wording does not mean that the matter in question does not exist. URL accessibility and application status reflect only the page status on the date of the last check.

Table C1. Availability, licensing and ethics registry for representative datasets. Panel A of 3; the identifier column is repeated.

Dataset	Availability	License	Ethics/Consent	Anonymization
VIPeR [15]	Legacy public download URL referenced; original project page not reliably accessible	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
CUHK03 [16, 17]	Legacy official download page indexed; direct legacy page currently inaccessible at last check	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
Market-1501 [18]	Registration required	Provider usage terms; no standardised open license identified	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
MSMT17 [19]	Available upon request; signed release agreement required	MSMT17 Release Agreement; NELVT/PKU copyright	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).

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Table C1 (continued), panel A

Dataset	Availability	License	Ethics/Consent	Anonymization
RegDB [22]	Official legacy link currently unavailable	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
SYSU-MM01 [23]	Available upon request; signed release agreement required	Dataset Release Agreement	Explicit participant consent/privacy license reported	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
RGBNT201 [24]	Publicly downloadable	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
PRAI-1581 [25]	Publicly downloadable	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
AG-ReID.v2 [27]	Publicly downloadable	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Ethics committee approval reported; individual consent not explicitly stated	Face pixelation applied
CARGO [14]	Publicly downloadable	Academic-use dataset terms; copyright retained	N/A (synthetic)	N/A (synthetic)
CP2108 [4]	Publicly downloadable	Dataset usage license referenced by official repository; specific terms not exposed on project page	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
AT-USTC [28]	Available upon request; signed release agreement required	Dataset Release Agreement	Volunteer consent explicitly reported	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
MP-ReID [3]	Available upon request; signed agreement required	Dataset agreement required	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Facial mosaicking; original raw footage deleted
WHU-MARS [5]	Available upon request; signed agreement required	WHU-MARS Dataset Agreement	Institutional authorization reported; on-site notices posted	Automatic face mosaicking; personally identifiable metadata removed

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Table C1 (continued), panel A

Dataset	Availability	License	Ethics/Consent	Anonymization
CUHK01 [S1]	Legacy official download page indexed; direct legacy page currently inaccessible at last check	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
CUHK02 [S2]	Legacy official download page indexed; direct legacy page currently inaccessible at last check	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
GRID [42]	Publicly downloadable	Research-use terms; no standardised open license identified	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
DukeMTMC-reID [S3, S4]	Withdrawn by creators; official distribution unavailable	Legacy CC BY-NC-SA 4.0 license	Withdrawn following privacy concerns	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
MARS [29]	Publicly downloadable	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
iLIDS-VID [30]	Publicly downloadable	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
LLCM [53]	Available upon request; signed release agreement required	LLCM Dataset Release Agreement	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
HITSZ-VCIM [31]	Publicly downloadable	Dataset-specific academic-use license	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
PRCC [84]	Publicly downloadable	Academic-use dataset terms; copyright retained by iSEE/SYSU	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
LTCC [82]	Available upon request; signed Database Release Agreement required	LTCC Database Release Agreement; Fudan University copyright	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).

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Table C1 (continued), panel A

Dataset	Availability	License	Ethics/Consent	Anonymization
VC-Clothes [S5]	Publicly downloadable	Apache License 2.0	N/A (synthetic)	N/A (synthetic)
DeepChange [83]	Available upon request; signed Dataset Access Agreement and institutional email required	DeepChange Dataset Access Agreement	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Face blocking required for dissemination; release-wide anonymization not stated
AG-ReID.v1 [61]	Publicly downloadable	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
AG-VPreID [32]	Publicly downloadable through official Kaggle/Google Drive links	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
CUHK-PEDES [34]	Available upon request; university email required	No public open license; access restricted due to licensing issues	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Not explicitly reported in the listed evidence sources (checked 2026-09-04).
RSTPReid [35]	Publicly downloadable	Repository is MIT-licensed; no separate dataset-specific license statement identified	Not explicitly reported in the listed evidence sources (checked 2026-09-04); images are derived from MSMT17.	Not explicitly reported in the listed evidence sources (checked 2026-09-04).

Table C1. Availability, licensing and ethics registry for representative datasets. Panel B of 3; the identifier column is repeated.

Dataset	Usage restriction	Maintenance	Availability checked
VIPeR [15]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Legacy resource; no explicit maintenance plan	2026-09-04
CUHK03 [16, 17]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Current CUHK dataset index available; legacy detail page unstable	2026-09-04
Market-1501 [18]	Academic/research use only; no redistribution; no commercial use	Legacy access page available; no explicit maintenance plan	2026-09-04

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Table C1 (continued), panel B

Dataset	Usage restriction	Maintenance	Availability checked
MSMT17 [19]	Academic/scientific research only; no redistribution; no commercial use; dataset images may not be published or shown	Official PKU dataset page and release agreement available; no explicit maintenance plan	2026-09-04
RegDB [22]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Legacy official page; unavailable at last check	2026-09-04
SYSU-MM01 [23]	Scientific research use; subject to release agreement	Official repository available; no explicit maintenance plan	2026-09-04
RGBNT201 [24]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Author dataset page and download link available; no explicit maintenance plan	2026-09-04
PRAI-1581 [25]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Official repository available; no explicit maintenance plan	2026-09-04
AG-ReID.v2 [27]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Official repository available; no explicit maintenance plan	2026-09-04
CARGO [14]	Academic purposes only; no commercial use	Official repository available; no explicit maintenance plan	2026-09-04
CP2108 [4]	Public study use indicated; no more specific restriction verified	Official repository actively updated; direct dataset link available	2026-09-04
AT-USTC [28]	Subject to dataset release agreement	Official repository available; no explicit maintenance plan	2026-09-04
MP-ReID [3]	Subject to dataset agreement	Official project page and repository available; no explicit maintenance plan	2026-09-04
WHU-MARS [5]	Non-commercial academic research only; redistribution prohibited	Official repository available; current 2026 release	2026-09-04
CUHK01 [S1]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Current CUHK dataset index available; legacy detail page unstable	2026-09-04
CUHK02 [S2]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Current CUHK dataset index available; legacy detail page unstable	2026-09-04
GRID [42]	Research purposes only; no commercial use	Official legacy dataset page available; no explicit maintenance plan	2026-09-04

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Table C1 (continued), panel B

Dataset	Usage restriction	Maintenance	Availability checked
DukeMTMC-reID [S3, S4]	Dataset withdrawn; current CVF ethics guidance treats it as unsuitable for submissions	Withdrawn; no active official distribution	2026-09-04
MARS [29]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Official legacy project/download page available; no explicit maintenance plan	2026-09-04
iLIDS-VID [30]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Official legacy dataset page available; no explicit maintenance plan	2026-09-04
LLCM [53]	Subject to dataset release agreement	Official repository and release agreement available	2026-09-04
HITSZ-VCM [31]	Academic use only; no redistribution; no commercial model use; trained models may only be distributed for non-commercial use	Official repository and download links available; no explicit maintenance plan	2026-09-04
PRCC [84]	Academic purposes only; no commercial use	Official project page and download links available; no explicit maintenance plan	2026-09-04
LTCC [82]	Research only; no commercial use; no redistribution, modification, publication, copying or dissemination of dataset records	Official project page and release protocol available; no explicit maintenance plan	2026-09-04
VC-Clothes [S5]	Use permitted under Apache License 2.0 terms	Official legacy project page and download links available; no explicit maintenance plan	2026-09-04
DeepChange [83]	Non-commercial research only; no redistribution; no commercial exploitation; faces must be blocked in disseminated images	Official project repository and access agreement available; no explicit maintenance plan	2026-09-04
AG-ReID.v1 [61]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Official repository and download link available; superseded by v2 but retained	2026-09-04
AG-VPRID [32]	Not explicitly reported in the listed evidence sources (checked 2026-09-04).	Official repository and competition materials available; FAQ intended for continued updates	2026-09-04
CUHK-PEDES [34]	Academic use only	Official project repository available; legacy project	2026-09-04

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Table C1 (continued), panel B

Dataset	Usage restriction	Maintenance	Availability checked
RSTPReid [35]	Underlying MSMT17 source-data restrictions require clarification	Official dataset repository and download links available; no explicit maintenance plan	2026-09-04

Table C1. Availability, licensing and ethics registry for representative datasets. Panel C of 3; the identifier column is repeated.

Dataset	Evidence source	Official URL or archived URL	Source version/date
VIPeR [15]	Original paper [15]; legacy Duke project page; Internet Archive index	https://web.archive.org/web/*/https://vision.cs.duke.edu/projects/viper/	Paper: 2008; archived landing-page index checked 2026-09-04
CUHK03 [16, 17]	Original paper [16]; CUHK person-identification data index; CUHK03-NP [17]	https://www.ee.cuhk.edu.hk/~xgwan/CUHK_identification.html	Papers: 2014/2017; page checked 2026-09-04
Market-1501 [18]	Original paper [18]; official Market-1501 project/access page	https://zheng-lab-anu.github.io/Project/project_reid.html	Paper: 2015; page checked 2026-09-04
MSMT17 [19]	Original paper [19]; official PKU VMC release page and MSMT17 Release Agreement	https://www.pkumc.com/	Paper: 2018; agreement/page state checked 2026-09-04
RegDB [22]	Original paper [22]; legacy project link referenced by the release	https://doi.org/10.3390/s17030605	Paper: 2017; legacy-link status checked 2026-09-04
SYSU-MM01 [23]	Original paper [23]; official SYSU-MM01 repository and Dataset Release Agreement	https://github.com/wuancong/SYSU-MM01	Paper: 2017; repository/agreement state checked 2026-09-04
RGBNT201 [24]	Original paper [24]; author dataset page/download link	No durable official or archived URL retained in the current evidence package; see [24].	Paper: 2021; source package checked 2026-09-04
PRAI-1581 [25]	Original paper [25]; official PRAI-1581 repository	https://github.com/stormyoung/PRAI-1581	Paper: 2021; repository state checked 2026-09-04
AG-ReID.v2 [27]	Original paper [27]; official AG-ReID.v2 repository	https://github.com/huynghuyen792/AG-ReID.v2	Paper: 2024; repository state checked 2026-09-04

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Table C1 (continued), panel C

Dataset	Evidence source	Official URL or archived URL	Source version/ date
CARGO [14]	Original paper [14]; authors' official project/repository	No durable official or archived URL retained in the current evidence package; see [14].	Paper: 2024; source package checked 2026-09-04
CP2108 [4]	Original paper [4]; official repository and dataset-usage-license reference	No durable official or archived URL retained in the current evidence package; see [4].	Original release; repository state checked 2026-09-04
AT-USTC [28]	Original paper [28]; official repository and Dataset Release Agreement	No durable official or archived URL retained in the current evidence package; see [28].	Original release; repository/agreement state checked 2026-09-04
MP-ReID [3]	Original paper [3]; official MP-ReID repository and dataset agreement	https://github.com/MP-ReID/mp-reid	Paper: 2025; repository/agreement state checked 2026-09-04
WHU-MARS [5]	Original paper [5]; official WHU-MARS repository and Dataset Agreement	https://github.com/msm8976/WHU-MARS	Release: 2026; repository/agreement state checked 2026-09-04
CUHK01 [S1]	Original paper [S1]; CUHK person-identification data index	https://www.ee.cuhk.edu.hk/~xgwan/CUHK_identification.html	Paper: 2012; page checked 2026-09-04
CUHK02 [S2]	Original paper [S2]; CUHK person-identification data index	https://www.ee.cuhk.edu.hk/~xgwan/CUHK_identification.html	Paper: 2013; page checked 2026-09-04
GRID [42]	Original paper [42]; legacy official GRID page	No durable official or archived URL retained in the current evidence package; see [42].	Original release; legacy-page status checked 2026-09-04
DukeMTMC-reID [S3, S4]	Original paper [S4]; creators' withdrawal notice; CVF ethics guidance	https://www.cv-foundation.org/ethics	Original release: 2016; withdrawal guidance checked 2026-09-04
MARS [29]	Original paper [29]; official MARS project/download page	https://zheng-lab-anu.github.io/Project/project_mars.html	Paper: 2016; page checked 2026-09-04
iLIDS-VID [30]	Original paper [30]; legacy iLIDS-VID dataset page	No durable official or archived URL retained in the current evidence package; see [30].	Paper: 2016; legacy-page status checked 2026-09-04

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Table C1 (continued), panel C

Dataset	Evidence source	Official URL or archived URL	Source version/ date
LLCM [53]	Original paper [53]; official LLCM repository and Dataset Release Agreement	https://github.com/ZYK100/LLCM	Paper: 2023; repository/agreement state checked 2026-09-04
HITSZ-VCIM [31]	Original paper [31]; official repository and dataset-specific license	No durable official or archived URL retained in the current evidence package; see [31].	Paper: 2022; repository/license state checked 2026-09-04
PRCC [84]	Original paper [84]; official project/download page	No durable official or archived URL retained in the current evidence package; see [84].	Paper: 2021; project-state checked 2026-09-04
LTCC [82]	Original paper [82]; official project page and LTCC Database Release Agreement	No durable official or archived URL retained in the current evidence package; see [82].	Paper: 2020; project/agreement state checked 2026-09-04
VC-Clothes [S5]	Original paper [S5]; official legacy project/download page	No durable official or archived URL retained in the current evidence package; see [S5].	Paper: 2020; project-page state checked 2026-09-04
DeepChange [83]	Original paper [83]; official project repository and Dataset Access Agreement	No durable official or archived URL retained in the current evidence package; see [83].	Paper: 2023; repository/agreement state checked 2026-09-04
AG-ReID.v1 [61]	Original paper [61]; official repository and download link	No durable official or archived URL retained in the current evidence package; see [61].	Original release; repository state checked 2026-09-04
AG-VPreID [32]	Original paper [32]; official repository and competition materials	No durable official or archived URL retained in the current evidence package; see [32].	Original release; repository/FAQ state checked 2026-09-04
CUHK-PEDES [34]	Original paper [34]; official project repository and access instructions	No durable official or archived URL retained in the current evidence package; see [34].	Original release; repository/access state checked 2026-09-04
RSTPReid [35]	Original paper [35]; official dataset repository; MSMT17 source-data terms	No durable official or archived URL retained in the current evidence package; see [35] and [19].	Original release; repository/source-data terms checked 2026-09-04

Note: RSTPReid is built on MSMT17. The MIT licence in its official repository ap-

plies to the repository code and does not automatically extend to images derived from MSMT17. The table’s evidence column lists both the underlying MSMT17 source and the access points traceable within the current evidence package. Use should be governed by the agreement for the underlying MSMT17 dataset.

Appendix D The twelve predefined query–gallery directions in MP-ReID

Table D1 expands the three factor-based groups in Section 4.1 of the main text into twelve specific directions to facilitate checking forward and reverse tasks. GR, GI, UR and UT denote Ground RGB, Ground Infrared, UAV RGB and UAV TIR, respectively.

Table D1. The twelve predefined query–gallery directions in MP-ReID.

No.	Setting category	Query→Gallery	Does the modality change?	Does the platform change?
1 [3]	Platform change only	UR→GR	No	Yes
2 [3]	Platform change only	GR→UR	No	Yes
3 [3]	Modality change only	GI→GR	Yes	No
4 [3]	Modality change only	GR→GI	Yes	No
5 [3]	Modality change only	UT→UR	Yes	No
6 [3]	Modality change only	UR→UT	Yes	No
7 [3]	Simultaneous modality and platform changes	UR→GI	Yes	Yes
8 [3]	Simultaneous modality and platform changes	GI→UR	Yes	Yes
9 [3]	Simultaneous modality and platform changes	UT→GR	Yes	Yes
10 [3]	Simultaneous modality and platform changes	GR→UT	Yes	Yes
11 [3]	Simultaneous modality and platform changes	UT→GI	Yes	Yes
12 [3]	Simultaneous modality and platform changes	GI→UT	Yes	Yes

Appendix E Structured ReID Protocol Registry

This appendix answers the question of exactly how the query and gallery are constructed. It neither replaces the data-content records in Appendix A nor automatically

combines multiple fixed source pairs into a single mixed-gallery evaluation.

Table E1. Structured ReID Protocol Registry. Panel A of 2; the identifier column is repeated.

Dataset / version	Training split	Test split	Query	Gallery
CUHK03 Original [16]	Original	Original	Manual / detected image	Manual / detected image
CUHK03-NP [17]	767	700	1,400 queries	5,328 / 5,332
Market-1501 [18]	751	750	3,368	15,913
RegDB [22]	206	206	Visible or thermal	Opposite modality
SYSU All [23]	296+99	96	IR	RGB
SYSU Indoor [23]	296+99	96	IR	RGB indoor
AG-ReID.v2 [27]	807	808	A, C or W	Specified target platform
CP2108 ALL [4]	1,442	666	Known source	Aerial + ground RGB
CP2108 A↔G [4]	1,442	666	Aerial / ground	Opposite platform
CARGO [14]	2,500 IDs / 51,451 images	2,500 IDs / 51,024 images	Aerial / ground source-specific	Opposite platform in fixed A↔G; mixed platform in ALL
AT-USTC [28]	135 IDs	135 IDs	RGB / IR	Ground (8 RGB + 8 IR cameras)
MP-ReID [3]	Pair-specific	Pair-specific	GR, GI, UR, UT	Pair-specific
WHU-MARS-2337 (AS-ReID) [5]	1,000 IDs / 217,504 images	1,337 IDs / 217,116 gallery images; 15,020 query images	Any known scenario	All known scenarios
WHU-MARS-1000 (AS-ReID) [5]	500 IDs / 92,313 images	500 IDs / 93,609 gallery images; 6,405 query images	Any known scenario	All known scenarios

Table E1. Structured ReID Protocol Registry. Panel B of 2; the identifier column is repeated.

Dataset / version	Direction	Mixed-source?	Repeats	Same-camera rule	Evidence boundary
CUHK03 Original [16]	Cross-camera	No	Protocol-dependent	Protocol-dependent	Original release statistics
CUHK03-NP [17]	Fixed NP	No	Fixed split	Junk same-camera positives	NP protocol only
Market-1501 [18]	Cross-camera	No	Fixed split	Same-camera junk filtered	Detected-box benchmark
RegDB [22]	V→T; T→V	No	10 trials	N/A / protocol-specific	Cross-spectrum pair
SYSU All [23]	All-search	No	Single / multi-shot	Indoor/outdoor rules	Filtered evaluation subset
SYSU Indoor [23]	Indoor-search	No	Single / multi-shot	Indoor camera rules	Filtered evaluation subset
AG-ReID.v2 [27]	A→C; A→W; C→A; W→A	No	Fixed split	Platform-specific	No mixed gallery
CP2108 ALL [4]	ALL	Yes	Fixed split	Protocol-specific	Mixed-platform RGB gallery
CP2108 A↔G [4]	A→G; G→A	No	Fixed split	Protocol-specific	Fixed direction only
CARGO [14]	ALL; G↔G; A↔A; A↔G	No (fixed A↔G); Yes (ALL)	Protocol-specific	Protocol-specific	Fixed A↔G does not establish universal mixed-gallery ability; ALL supports only its reported mixed-platform scope
AT-USTC [28]	Six Anytime-ReID scenarios	No (scenario-specific; mixed gallery not established)	Scenario-specific	Protocol-specific	Supports spectrum-time/clothing/anytime evaluation; does not by itself establish mixed-gallery retrieval
MP-ReID [3]	12 ordered directions: 2 platform-only, 4 modality-only, 6 joint	No	Pair-specific	Pair-specific	No joint split implied
WHU-MARS-2337 (AS-ReID) [5]	Any-scenario	Yes	Fixed benchmark	Closed-world identity	Known-source heterogeneous mixed gallery
WHU-MARS-1000 (AS-ReID) [5]	Any-scenario	Yes	Fixed benchmark	Closed-world identity	Known-source closed-world heterogeneous mixed gallery; subset of WHU-MARS-2337 split

Appendix F Task difficulties and methodological responses

This detailed mechanism table complements Figure 3 and Table 4 in the main text.

Table F1. Tasks and methodological responses.

Task setting	Main difficulties	Modelling response	Representative methods
Ground RGB	Limited samples, detection errors, large galleries	Metric learning; local–global; Transformer	XQDA; PCB; BoT; AGW; TransReID
Fixed cross-spectral settings	Spectral response and identity inconsistency	Dual-stream sharing; modality alignment; identity-level constraints	DDAG; CAJ; DEEN; SAAI / CSDN
Same-platform UAV settings	Top-down views, rotation, small scale	Rotation invariance; multi-scale; local structure	RotTrans
Aerial–ground	Viewpoint geometry and directional asymmetry	View disentanglement; local alignment; semantic assistance	VDT; VIF-AGReID; ViSA; SVPR-ReID
Fixed multi-modal, multi-platform settings	Joint spectral and platform variation	Scene conditioning; shared and source-specific representations; factor constraints	Uni-Prompt; CAJ / DEEN baselines
Multi-source mixed gallery	Unified ranking of cross-source candidates	Centre alignment; global identity prototypes	UAD; BoT / TransReID comparisons
Text/vision–language-related	Semantic priors and fine-grained identity information	Vision–language pretraining; attribute / viewpoint prompts	CLIP-ReID; SVPR-ReID
Domain adaptation/domain generalisation	Source–target domain shift	Pseudo-labels; clustering; representations for domain generalisation	MMT; SpCL; MetaBIN

Appendix G Detailed future evaluation settings

These settings are proposals, not established experiments. Table 5 in the main text gives their compact organisation. Identity supervision, combination visibility and gallery organisation remain separate reporting axes.

Table G1. Future multi-source supervision and open evaluation settings.

Evaluation setting	Training supervision	Test combination visibility	Gallery organisation	Main capability tested	Minimum evidence
Fully supervised multi-source	All known sources labelled	Seen combinations	Fixed or known mixed gallery	Known-source upper bound	Same model; source-level and overall results
Partial-source labels	Partial-source or partial-ID labels	Seen combinations	Known mixed gallery	Cross-source label propagation	Label source/ratio; labelled vs. unlabelled
Source-labelled/target-unlabelled	Source labelled; targets unlabelled	Target combination specified	Fixed or mixed target gallery	Multi-target adaptation	Source-target factors and metadata
Fully unsupervised multi-source	No ID labels	Seen combinations	Multi-source mixed gallery	Unsupervised unified identity space	Cluster purity; source bias; external prior
Held-out modality-platform combination	GR, UR and UT labelled	GI held out	Seen and held-out query sets; disjoint IDs	Factor recombination	Rank-1 and mAP per set; fixed-direction reference
Open-identity retrieval	Supervision range specified	Seen or held-out labelled	Gallery may omit query IDs	Unknown-identity rejection	Retrieval + rejection metrics; known/unknown ratio
Dynamic-gallery evaluation	Initial and later supervision specified	Continual source arrival	Dynamic open mixed gallery	Enrollment and continual adaptation	Temporal order; old/new IDs; update cost/forgetting

The GI-held-out example requires a common identity split across GR, GI, UR and UT. A factor-recombination interpretation also requires documenting GI/UT spectral equivalence; otherwise the result includes sensor novelty. Label-efficient variants must separately specify which identity annotations and source metadata are available.

Supplementary References

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